HW 3: A Two-Dimensional Hammer Throw with Air Resistance

Week 3 — Moodle exercise

Submission: on the course Moodle page, in the exercise that already contains the starting code.

Homework: a hammer throw with quadratic drag

In this exercise we practise a numerical simulation of a two-dimensional hammer throw from the origin ($x_0 = 0,\ y_0 = 0$) under gravity and quadratic air resistance (drag).

The equations of motion per unit mass are

$$\begin{aligned} \dot{x} &= v_x, & \dot{y} &= v_y, \\ \dot{v}_x &= -b\, v\, v_x, & \dot{v}_y &= -g - b\, v\, v_y, \end{aligned}$$

where $v = \sqrt{v_x^2 + v_y^2}$ is the instantaneous speed, $g = 9.81$ m/s^2, and $b \approx 0.001$ m^{-1} is the drag coefficient. We can solve these equations in the following way:

$$\begin{aligned} x_{n+1} &= x_n + v_{x,n}\,\Delta t, \\ y_{n+1} &= y_n + v_{y,n}\,\Delta t, \\ v_{x,n+1} &= v_{x,n} - b\, v_n\, v_{x,n}\,\Delta t, \\ v_{y,n+1} &= v_{y,n} - (b\, v_n\, v_{y,n} + g)\,\Delta t. \end{aligned}$$

Task (the Moodle exercise)

On Moodle you are given the beginning of a code that receives the following parameters:

- initial speed v_0 (in m/s);
- launch angle θ (in degrees);
- total simulation time T (in seconds);
- size of the time step Δt (in seconds);
- the drag coefficient b (in units of one over metre).

Solve the equations of motion with the scheme above until the moment the ball hits the ground ($y \le 0$) or the simulation ends ($t = T$), and print the distance the ball has travelled.

Physics extension (not for submission)

In your mechanics course you learned that to throw the ball as far as possible you should choose an initial angle of $45°$. Use $b = 0.001$ and check whether this result still holds. It is worth checking the distance for angles between $40°$ and $50°$.

In addition, you can solve the equations also for $b = 0$ and use `Plots.jl` to compare the trajectories of the two balls (`plot(x,y)`). How does adding friction affect the trajectory of the motion?